

Midazolam

(Versed)

- I. **Classification** – Benzodiazepine (Short Acting)
- II. **Action(s)**
 - GABA Receptor Stimulation causing CNS inhibition and CNS depression
- III. **Indication(s)**
 - Agitation / Excited delirium
 - Seizures
 - Cocaine or Amphetamine induced Tachycardia
 - Alcohol withdrawal
 - Shivering in Environmental Hyperthermia
 - Post-Intubation Sedation
- IV. **Contraindications**
 - Known hypersensitivity
 - Acute narrow-angle glaucoma
- V. **Precautions**
 - Monitor etCO₂, BP, SaO₂ prior to and post administration
 - Use caution in liver failure patients – may cause increased and prolonged effect
 - Caution if patient had hypotension or hypoventilation
 - GIVE SLOWLY (over at least 2 minutes for adults, longer for children and elderly)
- VI. **Drug interactions**
 - Benzodiazepines and narcotics – additive effect (respiratory depression/hypotension)
- VII. **Adult Dosing and Administration**
 - Seizure
 - 10 mg IM / IN
 - 5mg IV/IO
 - Repeat in 5 min to max total dose 10mg
 - Anxiety
 - 2mg IM/IN/IV/IO
 - Shivering in Environmental Hyperthermia
 - 5mg IM/IN
 - 2.5mg IV/IO
 - Repeat every 5 min to max total dose of 10mg to control shivering
 - Psychosis without immediate danger
 - 10mg IM
 - 5mg IV/IO
 - Repeat in 5 min to max total dose 10mg
 - Toxic Ingestion/Overdose of cocaine/amphetamine with agitation or significant tachycardia
 - 10mg IM
 - 5mg IV/IO
 - Repeat in 5 min to max total dose 10mg
 - Post Intubation Sedation
 - Versed 0.1mg/kg IO/IV (max 10mg) every 5-10 min titrated to sedation
 - Initial Versed bolus plus Versed 0.01-0.1mg/kg/hr (max 20mg/hr) continuous infusion
 - Call On-Call Medical Control Physician for additional doses

VIII. Pediatric Dosing and Administration

- Seizure/Toxic Ingestion with Cocaine/Amphetamine with agitation or significant tachycardia
 - 0.2 mg/kg IM / IN (max single dose 10 mg)
 - lowest volume acceptable for IN is 0.5mL
 - <25kg: 0.2mg/kg IM / IN
 - 25 kg – 49 kg: 5 mg IM / IN
 - $\geq$ 50 kg: 10 mg: 10 mg IM / IN
 - 0.1 mg/kg (max dose 5 mg) IV / IO
 - May titrate dose by 2 mg every 5 minutes (max total 10 mg)
- Post Intubation Sedation
 - Versed 0.1mg/kg IO/IV (max 10mg) every 5-10 min titrated to sedation
 - Initial Versed bolus plus Versed 0.01-0.1mg/kg/hr (max 20mg/hr) continuous infusion
- Call On-Call Medical Director for additional doses

IX. Onset

- 1-5 min

X. Duration

- 30 min - 5 hrs

XI. Possible Adverse Effects

- Respiratory depression
- Hypotension, Tachycardia
- Paradoxical CNS stimulation
- Commonly produces sedation, dizziness, fatigue